

# The matter dilution exponent is not identifiable from compressed CMB priors: a referent ambiguity, not a precision limit

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## Abstract

Consider the one-parameter family in which dark matter dilutes as  $\rho_{\text{dm}} \propto a^{-3+\epsilon}$  with a genuine cosmological constant. The Bianchi identity forces this background to be that of a fluid with constant equation of state  $w = -\epsilon/3$ , so the family is not an alternative to  $\Lambda$ CDM but a rewriting of it. Compressed CMB distance priors  $(\mathcal{R}, \ell_A, \omega_b)$ , together with the standard fitting formulas for  $r_d$ ,  $r_*$  and  $z_*$ , all require a *single* value of  $\omega_m$ . When  $\epsilon \neq 0$  there are two — the present-day label and the value at recombination — and nothing in the data selects one. We show on identical data (Pantheon+, DESI DR2 BAO, Planck 2018 distance priors, SH0ES) that this bookkeeping choice alone moves the recovered exponent from  $\epsilon = +0.0070$  to  $\epsilon = -0.0120$ , a range of 0.019, while every configuration quotes  $\sigma \simeq 0.002$ . We then show that this is *not* an information-loss effect: the compressed priors constrain  $\omega_c$  to  $\sigma = 0.0010$ , the same value the full Planck TT,TE,EE+low- $\ell$  likelihood delivers under identical fixed parameters, a ratio of 1.00 (bounded to ]0.5, 1.25[ by grid quantisation). The ambiguity is therefore one of *referent*, not of *precision*, and the spread is nineteen times the available precision. Consistently, published determinations of this exponent obtained with modified Boltzmann codes and full CMB likelihoods cluster at  $\epsilon = -0.00064 \pm 0.00045$  and span 0.005, while compressed-prior determinations span 0.016. We report one further trap: applying  $\epsilon$  to *total* matter rather than to dark matter alone creates an internal inconsistency worth  $7.5\text{--}15.1\sigma$  in  $\ell_A$ , because the same likelihood then treats baryons as standard in  $R_b$  and in the  $\omega_b$  prior. We claim no detection. We claim that a published  $\epsilon$  obtained from compressed priors carries an unquoted systematic of order  $10^{-2}$  unless the choice of  $\omega_m$  is stated.

## 1 The family, and why it has only one background

Write the dark sector as a matter component whose density obeys

$$\rho_{\text{dm}}(a) = \rho_{\text{dm},0} a^{-3+\epsilon}, \quad \rho_\Lambda = \text{const}, \quad \rho_b \propto a^{-3}, \quad \rho_r \propto a^{-4}. \quad (1)$$

It is tempting to read (1) as *pressureless matter that is not conserved*, the deficit being absorbed by nothing in particular. That reading is not available. With  $\Lambda$  genuinely constant,  $\nabla_\mu T^{\mu\nu} = 0$  applied to the total stress tensor forces the matter sector to carry a pressure

$$p_{\text{dm}} = -\frac{\epsilon}{3} \rho_{\text{dm}}, \quad \text{i.e.} \quad w_{\text{dm}} = -\epsilon/3. \quad (2)$$

“Non-conserved pressureless matter with constant  $\Lambda$ ” and “conserved matter with constant  $w = -\epsilon/3$  and constant  $\Lambda$ ” are the same model written twice. At background level there is one model, not two.

This is not new. Lima et al. (1996) give, for adiabatic matter creation with rate parameter  $\beta$ , the effective adiabatic index  $\gamma_* = \gamma(1 - \beta)$  and the scaling  $\rho = \rho_0(a_0/a)^{3\gamma(1-\beta)}$ , with creation

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pressure  $p_c = -\beta\gamma\rho$ . For dust ( $\gamma = 1$ ) this is exactly (1)–(2) with  $\epsilon = 3\beta$ . Two cautions, which we state because they are easy to lose: the identity  $\gamma_* = \gamma(1 - \beta)$  holds only for *constant*  $\beta$ , and the map  $\epsilon = 3\beta$  is *species-dependent* — for radiation ( $\gamma = 4/3$ ) the same  $\beta$  gives  $\rho_r \propto a^{-4(1-\beta)}$ . The map may not be carried across the radiation era.

We also note that the symbol  $\epsilon$  is used in a nearby but different literature — that of *decaying vacuum*, e.g. [Alcaniz & Lima \(2005\)](#) — where  $\Lambda$  is not constant. The family (1) is the constant- $\Lambda$  one, and at background level it coincides with the  $\Lambda$ wDM model constrained by [Kumar et al. \(2025\)](#).

## 2 Where the ambiguity enters

Compressed analyses of this family use three ingredients, and every one of them takes  $\omega_m$  as an input:

1. the shift parameter  $\mathcal{R} = \sqrt{\Omega_m H_0^2} D_A(z_*)/c$ , whose  $\sqrt{\Omega_m}$  prefactor is derived assuming  $\rho_m \propto a^{-3}$ ;
2. the fitting formulas for the drag and last-scattering sound horizons  $r_d, r_*$  ([Eisenstein & Hu, 1998](#)), calibrated on  $\Lambda$ CDM grids;
3. the fitting formula for  $z_*$  ([Hu & Sugiyama, 1996](#)).

In  $\Lambda$ CDM the input is unambiguous. In (1) it is not: the label  $\Omega_m$  and the density that actually governs pre-recombination physics differ by

$$\frac{\Omega_m^{\text{rec}}}{\Omega_m^{\text{label}}} = a_*^\epsilon, \quad a_* \simeq 1/1091, \quad \ln a_* \simeq -6.995. \quad (3)$$

For  $|\epsilon| = 0.005$  the two differ by 3.6%. An analyst must choose, and the literature does not say which. [Elgarøy & Multamäki \(2007\)](#) warned that  $\mathcal{R}$  “is not directly measurable from the cosmic microwave background, and its value is usually derived from the data assuming a spatially flat cosmology with dark matter and a cosmological constant”; we quantify what that warning costs in this family.

## 3 The measurement: four bookkeeping choices, one dataset

We use Pantheon+ (1580 SNe,  $z > 0.01$ , calibrators excluded), the 13 DESI DR2 BAO measurements, Planck 2018 distance priors, and the SH0ES  $H_0$  prior. Four configurations differ *only* in bookkeeping:

- (a) **label**: the fitting formulas and  $\mathcal{R}$  are fed the present-day  $\Omega_m$ ;
- (b) **coherent**: they are fed  $\Omega_m a_*^\epsilon$ , the density at recombination;
- (c) **direct, no  $\mathcal{R}$** :  $r_s$  is obtained by integrating  $\int c_s da/(a^2 H)$  in the model’s own background, and  $\mathcal{R}$  is dropped, leaving  $(\ell_A, \omega_b)$  with the marginal  $2 \times 2$  covariance;
- (d) **direct,  $\mathcal{R}$  coherent**: as (c) but retaining  $\mathcal{R}$  fed the recombination density.

The direct integrator is validated against the CAMB table it replaces: with  $N_{\text{eff}} = 3.046$  and the massive-neutrino contribution removed from the matter budget, the ratio  $r_s^{\text{direct}}/r_s^{\text{CAMB}}$  varies by  $1.3 \times 10^{-6}$  across the whole  $(\omega_b, \omega_m)$  grid, i.e.  $0.01\sigma$  in  $\ell_A$ . Its inferred drag redshift, recovered by inverting the integrator against CAMB’s  $r_{\text{drag}}$ , lands in  $[1057.8, 1062.1]$  against Planck’s 1059.9 — a check we did not impose.

| configuration                      | $\epsilon$    | $\sigma$ (opposable) | $\Delta\chi^2$ vs $\Lambda$ CDM |
|------------------------------------|---------------|----------------------|---------------------------------|
| (a) label                          | +0.0070       | 2.33                 | +9.36                           |
| (b) coherent                       | −0.0040       | 2.00                 | +9.36                           |
| (c) direct, no $\mathcal{R}$       | −0.0120       | 3.14                 | +9.87                           |
| (d) direct, $\mathcal{R}$ coherent | −0.0040       | 2.00                 | +9.90                           |
| <b>range</b>                       | <b>0.0190</b> |                      |                                 |

Table 1: Same data, same model, four bookkeeping choices. The quoted  $\sigma$  is the smaller of the profile half-width and the Wilks value  $\sqrt{\Delta\chi^2}$ ; where the two disagree the profile is non-parabolic and the local  $\sigma$  is not opposable.

Table 1 is the central measurement. The range is 0.019; each configuration quotes  $\sigma \simeq 0.002$ . Two remarks on honesty. First, configuration (c) reports a local significance of  $6.0\sigma$  that we do *not* quote: its profile is flat ( $\Delta\chi^2(2\sigma_{\text{local}})/4 = 0.39$  instead of 1) and the likelihood-ratio value is 3.14. Second, a decomposition of (c)’s gain shows 126% of it coming from the SH0ES term alone, with the supernovae worsening by 1.72 and the BAO by 0.96: that configuration buys  $H_0$ , it does not measure dilution. We report it because it belongs to the range, not because we believe it.

## 4 It is not information loss

The natural explanation — compressed priors carry only geometry, the full likelihood also carries the acoustic peak *heights* that pin  $\omega_m$  — is wrong, and we tested it rather than assert it.

The premise is correct: evaluating the compressed  $\chi^2$  at  $\ln(10^{10}A_s) = 3.000$  and 3.100 returns 2.575399113 both times, to machine precision. It carries no amplitude information whatsoever.

The conclusion does not follow. Profiling  $\omega_c$  with CAMB and the Planck TT,TE,EE+low- $\ell$  likelihood at fixed  $\theta_*$ ,  $\omega_b$ ,  $n_s$ ,  $\tau$ , and profiling  $\ln(10^{10}A_s)$  at each point, gives  $\sigma(\omega_c) = 0.0010$ . The same profile through the compressed priors gives  $\sigma(\omega_c) = 0.0010$ . The ratio is 1.00; propagating the grid quantisation bounds it in  $]0.5, 1.25[$ , far from any information-loss signature. As a consistency check, our  $\sigma$  is 1.20 times smaller than the marginalised Planck 2018 value  $\omega_c = 0.1200 \pm 0.0012$  (Planck Collaboration, 2020), which is what fixing four parameters should do; and the full  $\chi^2$  at its  $\Lambda$ CDM optimum reproduces our frozen anchor to 0.005.

So both determinations of  $\omega_m$  are equally sharp — 0.7% — and the spread across bookkeeping choices is 0.019, *nineteen times* the available precision.

The ambiguity is of referent, not of precision.

 (4)

$\mathcal{R}$  and the fitting formulas demand one  $\omega_m$ ; the family supplies two; the data adjudicate neither. Four analysts with identical data and identical 0.7% constraints obtain answers spanning 0.019 — not by noise, but by choice.

## 5 The baryon trap

A second failure mode is worth isolating because it is easy to fall into and it is large. If  $\epsilon$  is applied to *total* matter — i.e. if  $\Omega_m$  in (1) is read as baryons plus dark matter — then baryons dilute anomalously too. But the same likelihood typically (i) compares  $\omega_b$  to a prior measured under  $\rho_b \propto a^{-3}$  and (ii) integrates the sound horizon with  $R_b = (3\omega_b/4\omega_\gamma)a$ , a form that assumes the same. Consistency would require  $R_b \propto a^{1+\epsilon}$ . Measuring the difference:

Both exceed anything the analysis is trying to measure. The remedy is to restrict  $\epsilon$  to the dark sector, as Kumar et al. (2025) and Yao & Liu (2025) do. We note, against our own convenience, that the often-invoked argument “BBN forbids anomalous baryon dilution” is weaker than it sounds: requiring the BBN and CMB determinations of  $\omega_b$  to agree to  $\sim 2.5\%$  across the  $\simeq 12.5$

|                                                 | $ \epsilon  = 0.003$    | $ \epsilon  = 0.006$ |
|-------------------------------------------------|-------------------------|----------------------|
| $r_s$ under $R_b \propto a$ vs $a^{1+\epsilon}$ | $7.5\sigma$ in $\ell_A$ | $15.1\sigma$         |
| $\omega_b$ (recombination) vs the Planck prior  | $4.8\sigma$             | $8.1\sigma$          |

Table 2: Cost of applying  $\epsilon$  to total matter while treating baryons as standard elsewhere in the same likelihood.

| determination                                                    | method            | $\epsilon$      | $\sigma$       |
|------------------------------------------------------------------|-------------------|-----------------|----------------|
| <i>full likelihood, modified Boltzmann code</i>                  |                   |                 |                |
| Tsiapi & Basilakos (2019), $\Lambda(H)$ CDM2, Planck alone       | CAMB, Planck 2015 | +0.00148        | 0.00252        |
| Tsiapi & Basilakos (2019), $\Lambda(H)$ CDM2, joint              | CAMB, Planck 2015 | −0.00020        | 0.00181        |
| Tsiapi & Basilakos (2019), $\Lambda(H)$ CDM1, joint <sup>†</sup> | CAMB, Planck 2015 | +0.00302        | 0.00151        |
| Kumar et al. (2025), PL18+DESI DR2                               | CAMB, Planck 2018 | −0.00194        | 0.00096        |
| Li et al. (2025), Planck+DESI DR2+DESY5                          | CAMB, NPIPE       | −0.00126        | 0.00075        |
| Li et al. (2024), CMB+DESI DR1+DESY5 <sup>‡</sup>                | CosmoMC           | −0.00025        | 0.00092        |
| <b>weighted mean</b>                                             |                   | <b>−0.00064</b> | <b>0.00045</b> |
| range                                                            |                   | 0.0050          |                |
| <i>compressed distance priors</i>                                |                   |                 |                |
| Yang et al. (2025), five datasets                                | priors, DESI DR1  | −0.00612        | 0.00243        |
| this work, configuration (a)                                     | priors            | +0.0060         | 0.0030         |
| this work, configuration (b)                                     | priors            | −0.0030         | 0.0010         |
| this work, configuration (c)                                     | priors            | −0.0100         | 0.0020         |
| this work, direct $r_s$ , $\mathcal{R}$ label                    | priors            | +0.0020         | 0.0020         |
| range                                                            |                   | <b>0.0160</b>   |                |

Table 3: <sup>†</sup> Approximate conversion: this model’s  $Q$  is also sourced by baryons and radiation. We include it because it *widens* the full-likelihood family and therefore makes our own point harder to make. <sup>‡</sup> Nested predecessor of Li et al. (2025), same group; included and flagged. Conversions:  $\epsilon = 3\nu f$  or  $\epsilon = -3w_{\text{dm}}f$  with  $f = \rho_{\text{dm}}/\rho_m = 0.8389$ .

e-folds between the two epochs bounds only  $|\epsilon_b| \lesssim 2 \times 10^{-3}$ , and we could find no published constraint on the baryon dilution exponent itself. The defect above is one of *internal* inconsistency, not a violation of an external bound.

## 6 Confrontation with the published determinations

We collected every determination of this exponent we could verify at primary source, converting each into the convention of (1) by a declared rule, and split them by method.

The full-likelihood family is internally consistent ( $\chi^2/\text{dof} = 1.87$  for five degrees of freedom) and spans 0.0050; the compressed-prior family spans 0.0160, a factor 3.23. Excluding the flagged approximate conversion — the comfortable choice, which we do not take — the factor would be 4.67 and  $\chi^2/\text{dof} = 0.72$ .

Three caveats we grant in full. The six full-likelihood models are not identical to (1): Tsiapi & Basilakos (2019) let  $\rho_\Lambda$  run and carry a  $\Omega_{\text{dm}}/(1-\nu)$  prefactor in  $E^2$ , so at equal  $\epsilon$  their  $H(z)$  is not ours; Kumar et al. (2025) and Yao & Liu (2025) attach different perturbation prescriptions ( $c_s^2 = 0$  versus  $c_s^2 = w$ ) to the same background, and their intervals are not transferable to a background-only law. Their datasets differ from ours and from each other. And two of the six are the same group on nested data. Table 3 therefore measures a *dispersion*, and that is all we claim from it.

We also record, because it is the single most instructive item in the compressed column: [Yang et al. \(2025\)](#)’s headline  $\epsilon = -0.0073^{+0.0029}_{-0.0033}$  at  $2.4\sigma$  requires five datasets; on DESI+CMB+CC alone the same paper reports  $\epsilon = +0.0023^{+0.0055}_{-0.0067}$  and concludes there is no significant deviation. A sign reversal inside a single publication, on nested data.

## 7 What we do not claim

We do not measure  $\epsilon$ . None of our four configurations is a full likelihood, and the one that drops  $\mathcal{R}$  is driven by the  $H_0$  prior. We do not refute [Yang et al. \(2025\)](#) or anyone else: our pipeline, data and nuisance treatment differ. We do not claim that the field has abandoned  $\mathcal{R}$  — DESI DR2 conditions its baseline on  $(\theta_*, \omega_b, \omega_{bc})$  but explicitly tests the  $(\mathcal{R}, \ell_A, \omega_b)$  compression and reports it equivalent for the late-time models they consider ([DESI Collaboration, 2025](#)). Dropping  $\mathcal{R}$  is justified here narrowly, because  $\sqrt{\Omega_m}$  has no definition when matter does not dilute as  $a^{-3}$ .

We also disclose a systematic of our own. The distance priors we use are those of [Zhai & Wang \(2019\)](#), whose  $\ell_A = 301.80845$  corresponds to  $100\theta = 1.040923$ , i.e. the  $\theta_{\text{MC}}$  convention. Our  $\chi^2$  predicts  $\ell_A$  from CAMB’s exact  $z_*$  and  $r_*$ , i.e. in the  $\theta_*$  convention, yielding  $\ell_A = 301.75963$  and  $100\theta = 1.041091$  at the  $\Lambda$ CDM optimum. The two conventions differ by  $-0.54\sigma$  in  $\ell_A$  in every evaluation. This offset is common to all four configurations and therefore does not generate the spread of Table 1, but it is present and we state it rather than let a reader discover it.

## 8 Recommendations

1. A published constraint on a matter dilution exponent obtained from compressed CMB priors should state *which*  $\omega_m$  feeds  $\mathcal{R}$ ,  $r_d$ ,  $r_*$  and  $z_*$ , and should quote the systematic obtained by making the other choice. In this family that systematic is of order  $10^{-2}$  — an order of magnitude above the statistical error usually quoted.
2.  $\epsilon$  should be applied to the dark sector only, and the sound-horizon integrand should use the same baryon law as the background (§5).
3. Where the compressed route is kept,  $r_s$  should be integrated in the model’s own background rather than taken from a  $\Lambda$ CDM-calibrated fitting formula. This is standard practice in the early-dark-energy literature and costs nothing; we validate an implementation to  $1.3 \times 10^{-6}$  against CAMB in §3.
4. Finally, we record a gap. A systematic search of the matter-creation literature returned *no* analysis of any such model against full CMB power spectra: every one uses supernovae, BAO,  $H(z)$ , growth data, or a compressed CMB. The most recent example we found still describes its CMB input as “a compressed (also referred to as geometrical or background) CMB likelihood, where the CMB is treated as a BAO measurement at  $z \approx 1100$ ” ([Schiaivone et al., 2026](#)). Given §4, the cost of that choice is not a loss of precision — it is the introduction of a referent that the data cannot fix.

## Data and reproducibility

Every number in this paper is produced by a script whose success criteria were frozen by hash before execution, and re-verified afterwards by an independent tool that was itself mutation-tested (five injected faults, five detected). Where a frozen criterion later proved ill-posed, the criterion was left untouched and the defect recorded, rather than the criterion rewritten.

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